Quiz — Assignment 2

ECON 5783 — University of Arkansas

Name: _____

1. [2 points] Let

$$m(t) \equiv \mathbb{E}[\text{Exam Score}_i \mid \text{Tutoring Hours}_i = t].$$

In words, what does $m(4)$ represent? Describe how you could estimate it from a sample without imposing a linear relationship between tutoring hours and exam scores.

2. [2 points] An analyst models exam scores as a linear function of tutoring hours:

$$\text{Exam Score}_i = \alpha + \beta \text{Tutoring Hours}_i + u_i.$$

How does this model restrict the conditional expectation function compared with a fully flexible model?